The Elbow as a Deformable Solid System: Project Report

Mechanics of Solids · Nov 11, 2026 · @Rajendra

Introduction

This project treats the human elbow, meaning the biceps, the forearm bones and the joint between them, as a deformable solid system and analyses it with the tools of Mechanics of Solids. Like any engineering structure, the arm carries loads, deforms and develops internal stresses. Muscle, however, behaves very differently from the steel and concrete of a standard course: it is active, it carries tension only, and its stiffness depends on how strongly it is switched on.

The work is presented as an interactive website of eight linked pages. Each page isolates one idea, states its assumptions, derives the governing equations and lets the reader change the inputs and watch the stresses respond.

Objectives of the study

The study has four objectives, which move from the material, through the concepts, to a working model of the arm and its use.

1. **Explain the unusual behaviour of muscle as a deformable solid.** Muscle departs from the ideal linear elastic solid: it carries tension only, is nearly incompressible, is strongly anisotropic, and its stress depends on length, activation and time.
2. **Visually demonstrate the mechanics concepts of the muscle and elbow system.**
3. **Create a simplified heuristic of the biceps–elbow–forearm system.**
4. **Apply this heuristic under different loading conditions.**

Overview of the project

The website opens on a landing page of eight slides, read in order. The first three explain how muscle produces and grades force, the middle three apply solid mechanics to the muscle, elbow and forearm, and the last two apply the model and then test it against reality.

#	Page	What it shows	Main concepts
1	Objectives	The aim of the study and its four objectives	Scope of the study
2	Muscle anatomy and twitch	The muscle built up from proteins to the whole organ, and how a single twitch generates force at the sarcomere	Structure of an active material
3	Temporal summation and motor unit recruitment	How the nervous system grades force: overlapping twitches add up, and larger motor units are recruited in size order	Passive and active stress–strain curves; force = stress × PCSA
4	Muscle as a deformable solid	A loaded muscle in 3D cut by any plane, showing the traction vector, its normal and shear parts and the matching point on Mohr′s circle	Stress tensor, traction $t = \sigma \cdot n$, Mohr′s circle
5	Elbow system	The arm in static equilibrium as the elbow angle and the load change; the muscle must pull many times the weight in the hand	Moment balance, moment arm, joint reaction
6	Forearm as a beam	The forearm as a loaded beam, with the axial force, shear force and bending moment diagrams, and a 3D Mohr′s circle for any cut plane	Euler–Bernoulli beam theory, bending and shear stress
7	Curl variations	A standing curl and a preacher curl compared: the same dumbbell loads the biceps most at different muscle lengths	Applying the heuristic to two loading cases

#	Page	What it shows	Main concepts
8	Deviation from real life	Each simplification is removed in turn and replaced by the real anatomy, showing which way it moves the result	Limits of the model

Approach

Every model is quasi-static, so each state is an equilibrium and time-dependent effects are set aside. Each page lists its assumptions explicitly, typically uniaxial fibres, constant muscle volume, rigid or linear elastic bone, and a pinned elbow, so that the limits of the analysis are always visible.

Where possible, the constants come from the published literature rather than being invented:

- **Peak muscle stress:** 26.8 N/cm² (Persad et al. 2024)
- **Force–length curves:** active and passive shapes from Thelen (2003)
- **Motor unit sizes:** a 100-fold range (Fuglevand, Winter & Patla 1993)
- **Elbow flexor areas (PCSA):** cadaver data (Murray, Buchanan & Delp 2000)

Forearm dimensions are also taken from anthropometric sources. All the calculations run live in the browser, so every graph, stress value and Mohr's circle updates as the inputs change.

Conclusion

The project shows that the standard tools of Mechanics of Solids can be applied to a living joint. These are equilibrium, stress and strain, traction, Mohr's circle and beam theory. They explain why the biceps must pull many times the weight in the hand, and where in a curl the muscle and bone are most heavily stressed.

The final page is a deliberate check on these results. It shows that the real arm, with two forearm bones, several muscles, elastic tendons and viscoelastic tissue, departs from the simple model in predictable directions.